

① $n_{10} \rightarrow n_{16}$

$$\begin{array}{r}
 12345678 \mid 16 \\
 12345664 \mid 771604 \mid 16 \\
 \quad 14 \mid 771600 \mid 48225 \mid 16 \\
 \quad \downarrow \quad 4 \mid 48224 \mid 3014 \mid 16 \\
 \quad \quad \quad 1 \mid 3008 \mid 188 \mid 16 \\
 \quad \quad \quad \quad 6 \mid 176 \mid 11 \\
 \quad \quad \quad \quad \quad 12 \mid \downarrow \\
 \quad \quad \quad \quad \quad \quad \downarrow \quad B \\
 \quad \quad \quad \quad \quad \quad \quad C
 \end{array}$$

$12345678_{10} = BC614E_{16}$

$$\begin{array}{r}
 1000000 \mid 16 \\
 1000000 \mid 62500 \mid 16 \\
 \quad 0 \mid 62496 \mid 3906 \mid 16 \\
 \quad \quad 4 \mid 3904 \mid 244 \mid 16 \\
 \quad \quad \quad 2 \mid 240 \mid 15 \\
 \quad \quad \quad \quad 4 \mid \downarrow \\
 \quad \quad \quad \quad \quad F
 \end{array}$$

$1000000_{10} = F4240_{16}$

② $n_{16} \rightarrow n_{10}$

$$\begin{aligned}
 12345678_{16} &= 1 \cdot 16^7 + 2 \cdot 16^6 + 3 \cdot 16^5 + 4 \cdot 16^4 + 5 \cdot 16^3 + 6 \cdot 16^2 + 7 \cdot 16^1 + 8 \cdot 16^0 = \\
 &= 1 \cdot 268435456 + 2 \cdot 16777216 + 3 \cdot 1048576 + 4 \cdot 65536 + 5 \cdot 4096 + 6 \cdot 256 + 7 \cdot 16 + 8 \cdot 1 = \\
 &= 268435456 + 33554432 + 3145728 + 262144 + 20480 + 1536 + 112 + 8 = 305419896_{10}
 \end{aligned}$$

$12345678_{16} = 305419896_{10}$

$$1000000_{16} = 1 \cdot 16^6 + 0 \cdot 16^5 + 0 \cdot 16^4 + 0 \cdot 16^3 + 0 \cdot 16^2 + 0 \cdot 16^1 + 0 \cdot 16^0 = 1 \cdot 16777216$$

$1000000_{16} = 16777216_{10}$